

PORTOFOLIO PROJECT 1

SUPERSTORE SALES ANALYSIS

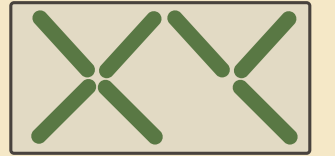
Identifying Profitability Drivers Through SQL-Based Analysis

sql · Bigquery

Looker Studio

Jan 2019 - Des 2022

BUSINESS PROBLEM



01

Which product & categories generate high sales but drain profit?

02

How do discount strategies impact overall profitability?

03

Which customers and regions drive - or destroy - business value?

Dataset & Approach

9,993

Total Rows

793

Unique Customers

1850

Unique Product

4 Years

Jan 2019 - Desc 2022

SQL Techniques Used

- Common Table Expressions (CTE)
- Windows Functions : RANK(), LAG()
- CASE WHEN for conditional segmentation
- Aggregation: SUM, COUNT, AVG, ROUND
- Month-over-month growth with LAG()
- Data Quality filtering via margin validation

Data Quality Note

~4,825 rows (48%) contained profit values inconsistent with their sales figures - a data integrity issue in the source dataset

These rows were excluded from all profit analyses. Only **5,169 validated rows** were used - identified using a margin validation filter ($|\text{profit/sales}| \leq 200 \%$)

KEY MATRICS

\$ 1.03 M

Total Sales
(Full Dataset)

10.4%

Overall Profit
Margin

793

Unique
Customers

5,009

Total Orders



1.850

unique product analyzed
across 3 categories



15.6%

of revenue from top 10
customers only



Sept 2021

peak sales month -
\$56M in a single month

Product Profitability

257

out of 1,29
product

generates
negative profit

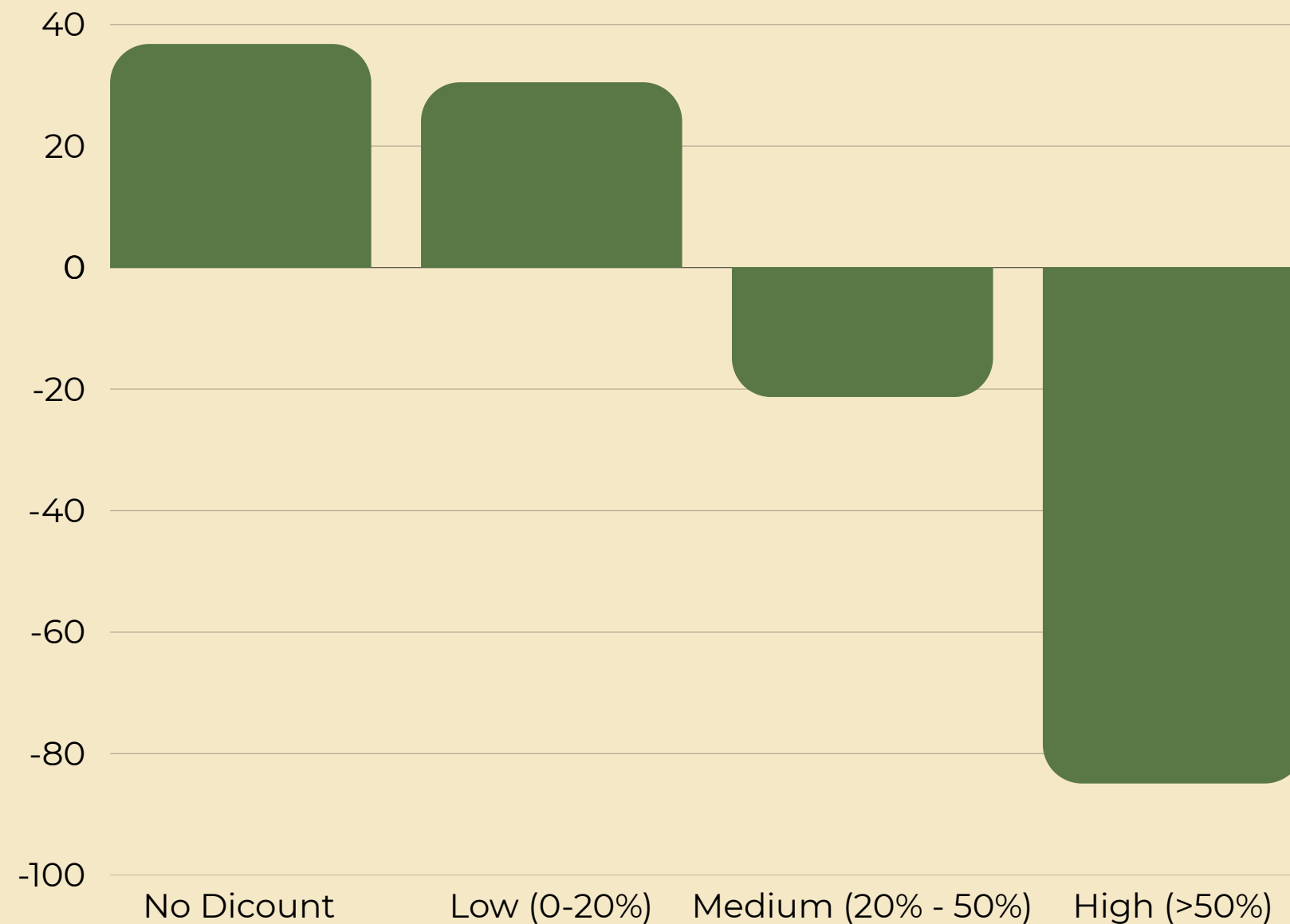
16.8%
of all product

Worst Performing Product by Loss

Product	Sales	Profit Loss	Margin
O’Sullivan Living Dimensions Bookcase	\$17.4M	-\$5.4M	-31.3%
O’Sullician Plantations 2-Door Library	\$23.5M	-\$5.6M	-23.9%
Bretford CR4500 Rectangular Table	\$14.8M	-\$2.9M	-19.8%
Atlantic Metals 5-shelf Bookcases	\$23.9M	-\$0.5M	-2.2%
Lexmark MX611dhe Laser Printer	\$16.8M	-\$1.2M	-7.1%

Insight : Furniture subcategory (Bookcases, Tables Dominates loss-making products. Revenues cannot be the sole performance metric - product-level profitability monitoring is essential.

Discount Impact Analysis



No Discount

+ 36.8%

2,038 transactions

Low (0-20%)

+ 30.5%

2,658 transactions

Medium (20 - 50%)

-21.3%

219 transactions

High (>50%)

-84.9%

214 transactions

Recomendation : Cap discount at 20%. Any discount above this threshold produces negative margins - a policy review is essential

Customer Segmentations

54%

Low Value

419 customers

Below avg order frequency
(< 6.6 orders)

27%

Medium Value

213 customers

6.6 - 9.9 orders
per customer

19%

High Value

149 customers

Above 1.5x avg frequency
(> 9.9 orders)

Top 5 Customers by Revenue

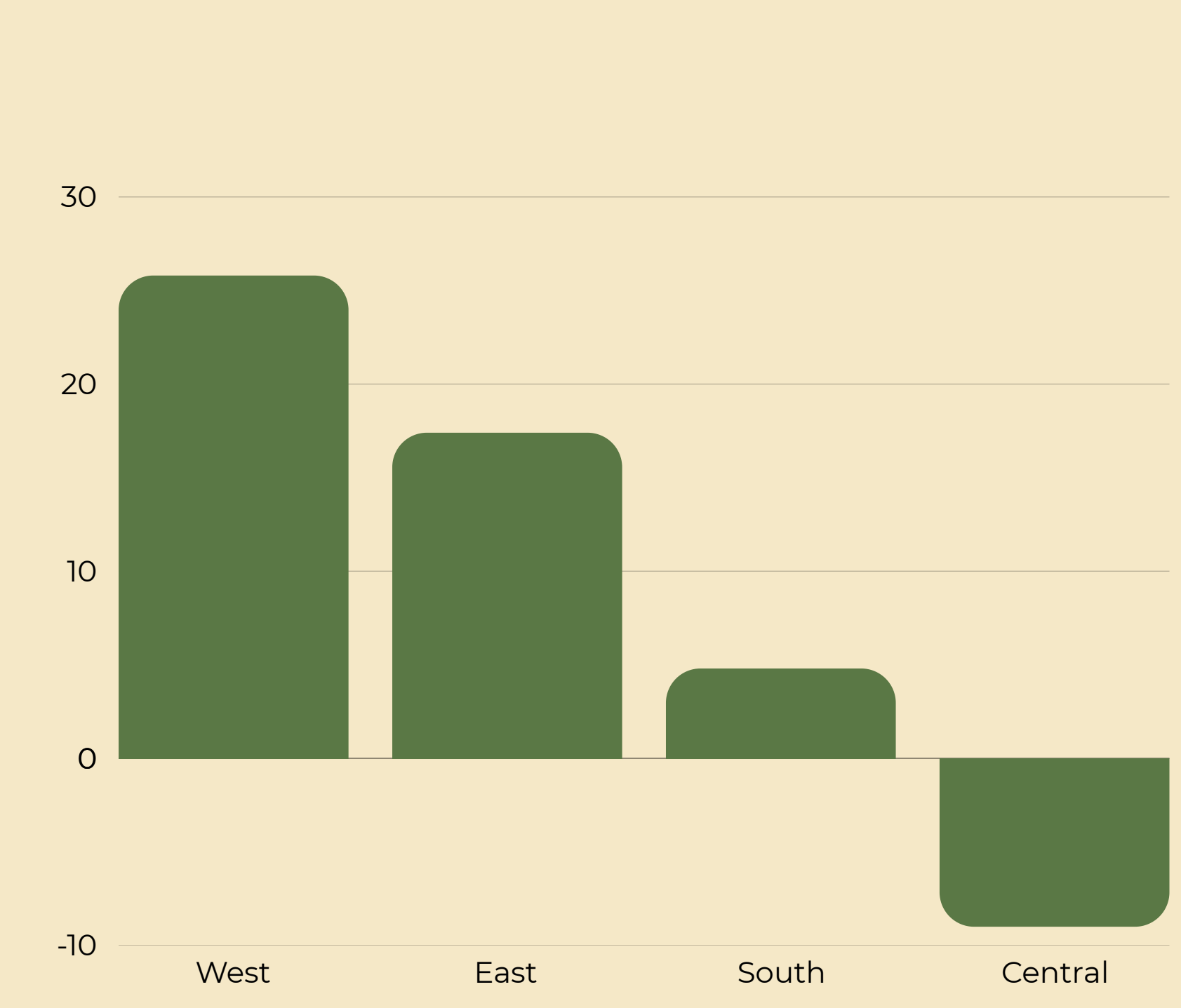
Customer	Total Sales	Orders
Anna Gayman	\$21.8M	11
Craig Molinary	\$14.8M	9
Amy Hunt	\$13.6M	6
Patrick O'Brill	\$11.5M	14
Emily Phan	\$10.5M	11

Key Insight

Top 10 Customers = 15.6%
iof total revenue

High frequency $\neq$ high spend
Emily Phan: 11 orders, \$10.5M
Patrick O'Brill: 14 Orders, \$11.5M

Regional Performance



West

+25.8%

\$284.3M Sales

East

+17.4%

\$153.5M Sales

South

+4.8%

\$94.8M Sales

Central

-9.0%

\$252.9M Sales

Central : high volume, negative margin - signals a structural pricing or cost issue, not a volume problem

BUSINESS RECOMMENDATIONS

01

Cap Discount at 20%

Medium & high discount produce negative margins (-21% and -85%). Require management approval for any discount above 20%

02

Retain Top 10 Customers

Top 10 customers = 15.6% of total revenue. A dedicated retention program for this group would have immediate ROI

03

Review Furniture Pricing

Tables (-38% margin) and Bookcases (-8%) are structural loss-makers. Renegotiate supplier costs or reprice these lines.

04

Audit Central Region

\$252M in sales but -9% margin. Investigate regional discounting behavior, pricing, and cost structure.

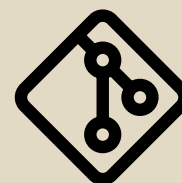
05

Invest in Technology Product

Copiers (+82%) and Phones (+63%) are the strongest margin drivers Prioritize marketing and stock investment here

THANK YOU

Reta Tri Asriani | Entry-level data analyst



<https://github.com/retatriasriyani/Superstore-Sales-Performance-Analysis-SQL-Looker-Studio.git>



<https://datastudio.google.com/reporting/651229bb-3218-411c-9bb7-f4e3788a29d9>